REFERENCE AGREEMENT IS NOT ACOUSTIC CORRECTNESS: EVALUATING LARGE AUDIO LANGUAGE MODEL CAPTIONS ON CLOTHO

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ABSTRACT

Audio captions are usually evaluated by their agreement with human references, yet a caption can disagree with the references because it describes audible content they omit, or agree with them while misdescribing the recording. We examine this gap for large audio language models (LALMs). Captions from GPT-Audio-1.5, Gemini 3.8 Flash, Qwen2.5-Omni-7B and Audio Flamingo 3 for the 1,045 recordings of the Clotho v2.1 evaluation split are scored with twelve reference-based and two audio-aware metrics and compared with a matched human leave-one-out baseline. Every model scores below the human baseline on every reference-based metric, and the two open-weight models score highest. A single-reviewer assessment of 200 sampled captions, first against the references and then against the audio, reverses this order: GPT-Audio-1.5, ranked third or fourth by the text metrics, was rated accurate for 47 of 50 recordings, and 56 of the 64 captions with weak reference agreement had no major acoustic error. Within recordings, no evaluated metric showed significantly above-chance concordance with the audio ratings, although the intervals are wide. Reference agreement is therefore insufficient on its own to establish acoustic correctness.

***Index Terms*—** Audio captioning, large audio language models, evaluation metrics, Clotho, human references

1. INTRODUCTION

Automated audio captioning (AAC) describes the sound sources, events and acoustic characteristics of a recording in natural language [1]. Large audio language models (LALMs) now perform this task without task-specific training, and produce longer and more detailed descriptions than captioning systems trained on AAC datasets [2, 3, 4, 5].

AAC systems are evaluated by comparing their captions with human references, using metrics such as BLEU [6], METEOR [7], CIDEr [8] and SPIDEr [9]. A recording often contains several sounds, and each listener describes a selection of them. Two cases then arise that the scores conflate: a caption can disagree with the references because it describes audible content that they omit, and a caption can agree with the references while misdescribing the recording. A reference-based score alone cannot tell whether low agreement marks an acoustic error or a legitimate description that the references do not cover. The problem is sharper for LALMs, whose captions differ from the references in length and style.

We study this problem for four LALMs on Clotho v2.1 [10] with three research questions, each building on the previous one:

1. **RQ1: How closely do model captions agree with the five human reference captions?** This establishes the degree of reference agreement under twelve reference-based and two audio-aware metrics.
2. **RQ2: How do model captions compare with held-out human captions in metric scores and caption length?** This calibrates that agreement against human captions scored against the same references.
3. **RQ3: What patterns of agreement, additional detail, omission and mismatch appear in model captions?** This relates reference agreement to acoustic correctness through a review of sampled captions against the references and then against the audio.

In the reviewed sample, the metrics follow the reference-only review, but we find no statistical evidence that they follow the audio-grounded ratings.

2. RELATED WORK

2.1. Automated audio captioning

AAC benchmarks such as AudioCaps [11] and Clotho [10] provide several crowdsourced captions per evaluation recording, and the DCASE challenge has made them standard [1]. In Clotho, annotators write captions, other annotators correct or rephrase them, and the five captions rated highest for accuracy and fluency are kept [12]. Each Clotho reference is a single sentence of 8–20 words, so it describes a selection of the sounds, typically the most salient, rather than the whole recording. Pre-training on larger, weakly labelled collections such as WavCaps [13] keeps trained systems close to this caption style. When a captioner describes more of a recording than its references do, the references no longer bound what a correct caption may contain.

2.2. Large audio language models

LALMs couple a pre-trained audio encoder with a large language model and follow free-form instructions about audio [14, 2, 3, 15]. Recent models are either general multimodal assistants, such as Qwen2.5-Omni [4], or audio specialists, such as Audio Flamingo 3 [5]. Unlike AAC systems trained on reference captions, they are not tuned to the length or vocabulary of a benchmark, so their captions can contain correct detail that no reference mentions. They can also describe sounds that are absent from the recording [16]. Deciding whether an added detail is correct or hallucinated therefore requires the audio, not the references.

2.3. Audio caption evaluation metrics

Reference-based metrics compare a caption with the human references. BLEU [6] and CIDEr [8] count n-gram overlap, METEOR [7] also matches stems and synonyms, SPICE [17] compares scene graphs, and SPIDEr [9] averages CIDEr and SPICE. Zhou et al. [18] found that these metrics agree poorly with human judgements of audio captions and proposed FENSE, which adds a fluency

penalty to Sentence-BERT similarity. Later metrics use contextual embeddings [19], sound descriptors [20], a language-model judge [21] or several candidates [22], but all measure similarity to the references, not to the recording. Audio-aware metrics use the recording itself: CLAP [23, 24] embeds audio and text in a joint space, analogous to CLIPScore [25] for images, and MACE [26] combines CLAP audio–text and text–text similarity with a fluency penalty. Both measure global similarity; neither verifies individual sound events.

2.4. Reference agreement and acoustic grounding

Interpreting a reference-based score requires answering two separate questions: how well humans agree with each other, and whether the caption is right about the recording. The first is answered by scoring human captions against the remaining references, as for COCO image captions [27]; such a baseline measures inter-reference agreement, not correctness. The second needs the signal itself, as in measures of object hallucination that check captions against the image [28]. AAC metric studies have compared scores with human preferences between caption pairs, judged on description accuracy and fluency [18, 26], but did not rate each caption on a categorical correctness rubric or assess the details that no reference mentions. We combine a matched human leave-one-out baseline with a manual audio-grounded assessment of sampled LALM captions and of their reference-unsupported phrases.

3. METHODOLOGY

3.1. Data and models

We use the complete evaluation split of Clotho v2.1 [10]: 1,045 recordings of 15–30 s (6.5 h in total), each with five human captions (5,225 captions). We evaluate two API models, GPT-Audio-1.5 and Gemini 3.8 Flash, and two open-weight models, Qwen2.5-Omni-7B [4] and Audio Flamingo 3 [5], at pinned checkpoint revisions. Every model received a single user turn with the full recording followed by the same instruction (prompt template below). No references were shown, no length was imposed, and no model-specific prompt tuning was done; only the provider-specific encoding of the audio differed.

Prompt template (one user turn, identical for all models)
`[audio]` *full recording, 15–30 s, no cropping*
`[text]` Describe the sounds heard in this audio recording in one clear sentence that mentions the main sound sources, audible events, and distinguishing acoustic characteristics. Do not speculate about sources or events that are not supported by the audio. Respond with the sentence only, as plain text, without JSON, lists, labels or other formatting.

Decoding used temperature 0, top-p 1 and a seed of 42 where supported; Gemini 3.8 Flash ran at its lowest thinking level (`low`), as thinking cannot be disabled. This gave one caption per model and recording (4,180 in total). Of GPT-Audio-1.5's captions, 85 were sentences wrapped in JSON and were unwrapped, and 21 malformed ones were scored as returned; excluding their recordings changes no model order and no significance decision. The API models are not deterministic at temperature 0, so all results are conditional on the released captions; checkpoint hashes, API run dates and library versions are listed in the repository.

3.2. Metrics and human leave-one-out calibration

Metrics. The twelve reference-based metrics are BLEU-1 to BLEU-4, METEOR, CIDEr-D, SPICE, SPIDEr, SPIDEr-FL, FENSE, and the raw and baseline-rescaled F1 of BERTScore, computed on PTB-tokenized text with the aac-metrics package [29]. The two audio-aware metrics are CLAP similarity, the cosine between the audio and caption embeddings of MS-CLAP 2023 [24] (each clip embedded whole as the duration-weighted mean of 7 s chunks), and MACE [26].

Matched human calibration. A model's score against the references is hard to interpret without knowing how well a human caption agrees with the other references. We therefore hold out each human caption h_j of a recording in turn and score it against the remaining four, $R_{-j} = H \setminus \{h_j\}$, where $H = \{h_1, \ldots, h_5\}$. Each model caption is scored against the same five four-reference subsets $R_{-1}, \ldots, R_{-5}$, so model and human captions face identical references. Scores are averaged over j per recording; corpus-level metrics are computed per j and averaged. This baseline measures inter-reference agreement, not caption quality or acoustic correctness. Because CIDEr-D, and hence SPIDEr and SPIDEr-FL, depend on the reference set, five-reference (RQ1) and four-reference (RQ2) scores are never compared with each other. Clotho's protocol keeps original and rephrased captions without deduplication, and 33% of held-out captions have a word-set Jaccard overlap of at least 0.6 (an operational threshold, not a test of paraphrase) with a reference; as a sensitivity analysis, we repeat RQ2 without them.

Reproducibility note. BERTScore (RoBERTa-large, last layer, no IDF) uses corrected batching: torchmetrics 1.9.0, called by aac-metrics, pairs batched captions with other items' references. The fix and a test are in the repository.

3.3. Caption length and qualitative review

Caption length is the number of whitespace-separated words; we relate it to each score with Spearman correlations over captions. For RQ3, we drew 50 recordings at random (seed 42), independently of the scores, and reviewed the four model captions of each, 200 in total. The review had two stages.

Stage 1, reference-only. Without audio, the reviewer compared each caption with the five references and assigned one of six reference-agreement categories: aligned (1 strong agreement, 2 mostly aligned, 3 mostly aligned with additional details), partially aligned (4 with missing details, 5 with additional details), or 6 weak agreement or source/event mismatch. Phrases that no reference mentions (*reference-unsupported phrases*) were listed.

Stage 2, audio-grounded. After the stage-1 labels were final, the reviewer listened to each recording and rated each caption: (1) accurate, where omitting a secondary sound alone does not lower the rating; (2) minor errors: principal source and event correct, but a wrong attribute or order or an unsupported secondary detail; or (3) major error: principal source or event wrong or invented, or the dominant sound omitted. Each reference-unsupported phrase was judged audible, inaudible or unverifiable in the recording.

One reviewer performed both stages by hand; no AI system assigned or suggested labels. Captions were blinded to model identity under per-recording random labels, but the reviewer had seen the references before listening.

3.4. Statistical analysis

The recording is the unit of analysis, and every comparison is paired. We test mean differences with two-sided sign-flip permutation tests

(10,000 permutations) and report 95% percentile bootstrap intervals over recordings (10,000 resamples). Holm correction is applied within predefined families over all 14 metrics: 84 model-pair tests for RQ1 and 56 model-versus-human tests for RQ2. We report the paired effect size d_z, because with 1,045 recordings very small differences are significant. Ranking stability is the share of bootstrap resamples that reproduce the observed model order. To relate a metric to the audio ratings within recordings, we take every pair of captions of the same recording with different ratings and compute the share in which the metric scores the better-rated caption higher, counting ties as one half (*concordance*; 0.5 is chance). Its intervals and tests resample or sign-flip whole recordings, with Holm correction over the ten metrics shown in Table 1. Review proportions carry Wilson 95% intervals.

4. RESULTS

4.1. RQ1: Agreement with five references

Audio Flamingo 3 scores highest and Qwen2.5-Omni-7B second on every reference-based metric, and the two API models take the third and fourth places (Table 1, left). The order is stable: at least 99% of bootstrap resamples reproduce it for nine of the twelve metrics, and only the order of the two API models varies for BLEU-3 (87%), SPIDEr and SPIDEr-FL (55%). After Holm correction, 76 of the 84 model-pair differences are significant.

The audio-aware metrics order the models differently: CLAP similarity and MACE score GPT-Audio-1.5 close to Qwen2.5-Omni-7B (0.466 and 0.462; 0.573 and 0.575), and Gemini 3.8 Flash last. Per caption, the text metrics correlate with each other (Spearman ρ = 0.41–0.88), and CLAP and MACE correlate 0.89 with each other but only 0.28–0.67 with the text metrics. These correlations describe how the metrics vary together, not which of them agrees with the audio.

4.2. RQ2: Comparison with held-out human captions

Under the same four references, every model scores below the held-out human captions on every reference-based metric, and all of these differences are Holm-significant (Table 1, right). Because the baseline measures how closely one human caption follows the other four, this shows that the models match Clotho's reference conventions less closely, not that their captions are less correct. The effect sizes (d_z from -0.48 to -2.55) are standardized differences in metric scores, not in caption quality.

The audio-aware metrics are the exception: on CLAP similarity, Audio Flamingo 3 is not significantly different from the human captions ($+0.006$, $d_z = 0.06$), the only one of the 56 comparisons that is not significant. Without near-paraphrase captions, the human baseline falls (CIDEr-D 0.91 to 0.46), but every model remains significantly below it on all reference-based metrics; Audio Flamingo 3 rises above it on CLAP similarity, at the uncorrected level only.

Model captions are 1.45 (Gemini 3.8 Flash) to 1.85 (GPT-Audio-1.5) times as long as human captions, and 92% of GPT-Audio-1.5's captions are longer than every reference of their recording. For the models, length is negatively associated with the overlap metrics, from -0.09 (Gemini 3.8 Flash) to -0.63 (Audio Flamingo 3) for CIDEr-D and from -0.19 to -0.52 for rescaled BERTScore. CLAP similarity and MACE are almost independent of length except for Audio Flamingo 3 (-0.11 and -0.17), and human captions show near-zero correlations throughout. These associations do not isolate a causal effect of length.

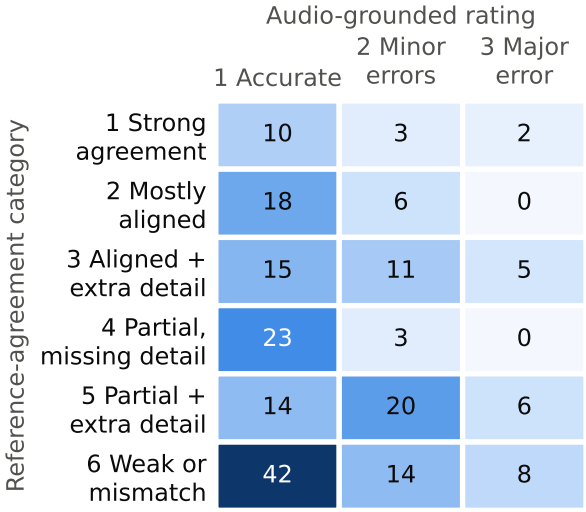


Fig. 1. Stage-1 reference-agreement category (rows; references only) by stage-2 audio-grounded rating (columns; by listening) for the 200 sampled captions of all four models, single reviewer.

4.3. RQ3: Agreement patterns and acoustic correctness

The sample was similar to the full set on the measured summary statistics (standardized differences of at most 0.37). *Two reviews.* Against the references, Audio Flamingo 3 and Qwen2.5-Omni-7B are most often aligned and Gemini 3.8 Flash least often, and GPT-Audio-1.5 has the most captions with missing detail (Table 2). Against the audio, the order reverses. GPT-Audio-1.5 was rated accurate for 47 of 50 recordings (94%, 95% CI 84–98%) with no major error, while the other models were rated accurate for 23 to 27 (46–54%, intervals within 33–67%) and had 5 to 9 major errors (single-reviewer ratings).

Details absent from the references. The 200 captions contain 262 reference-unsupported phrases. Of these, 64% were audible in the recording, 28% inaudible and 8% unverifiable (95% CI for audible 58–70%, too narrow because phrases of a caption are not independent). Inaudible phrases occur in 26 of Gemini 3.8 Flash's captions, 19 of Qwen2.5-Omni-7B's and 15 of Audio Flamingo 3's, but in only one of GPT-Audio-1.5's. At the caption level (Fig. 1), 56 of the 64 captions in category 6 had no major error (42 accurate, 14 minor errors), and 23 of the 26 category-4 captions were rated accurate. In this sample, weak reference agreement therefore did not imply an incorrect description. Incomplete reference coverage is a plausible explanation consistent with these results; reference errors, ambiguous sounds and reviewer judgement may also contribute.

Metrics and the two reviews. Across models, the metrics follow the reference-only review: Kendall's τ between each metric's model order and the reference-review order is 0.67–1.00, but with the audio-review order it is 0–0.33 for every metric except CLAP similarity (0.67). Mean scores fall from category 1 to category 6 on every metric, but are flat or reversed across audio ratings (rescaled BERTScore 0.29, 0.31 and 0.34 for ratings 1–3). Within recordings, across 156 caption pairs from 42 recordings, concordance with the audio ratings ranges from 0.39 (rescaled BERTScore) to 0.58 (MACE, 95% CI [0.46, 0.69]; CLAP similarity 0.56 [0.45, 0.66]), and no metric differs significantly from chance after Holm correc-

Table 1. Mean per-recording scores. RQ1 (left): model captions against all five references; bold marks the highest model. RQ2 (right): held-out human captions against the other four references, and the paired effect size d_z of each model minus that baseline under the same references; all Holm-significant except †. Five- and four-reference scores are not comparable. Last row: mean words, and d_z of model minus mean reference length. GPT: GPT-Audio-1.5; Gemini: Gemini 3.8 Flash; Qwen: Qwen2.5-Omni-7B; AF3: Audio Flamingo 3.

	RQ1: models, five references				RQ2: human	RQ2: model − human, d_z (four ref.)			
Metric	GPT	Gemini	Qwen	AF3	(four ref.)	GPT	Gemini	Qwen	AF3
BLEU-1	0.277	0.251	0.368	**0.407**	0.640	−2.49	−2.55	−1.67	−1.40
BLEU-4	0.004	0.002	0.015	**0.030**	0.234	−1.27	−1.28	−1.18	−1.05
METEOR	0.124	0.102	0.151	**0.174**	0.319	−1.84	−2.05	−1.57	−1.32
CIDEr-D	0.048	0.064	0.164	**0.191**	0.906	−1.83	−1.75	−1.54	−1.41
SPICE	0.067	0.051	0.091	**0.117**	0.232	−1.47	−1.58	−1.19	−0.91
SPIDEr	0.057	0.058	0.127	**0.154**	0.569	−1.90	−1.84	−1.59	−1.41
FENSE	0.401	0.387	0.474	**0.510**	0.575	−1.10	−1.05	−0.73	−0.48
BERTScore-F1 (rescaled)	0.254	0.238	0.331	**0.352**	0.598	−2.40	−2.47	−1.80	−1.51
CLAP similarity	0.466	0.381	0.462	**0.509**	0.503	−0.34	−0.94	−0.42	+0.06†
MACE	0.573	0.489	0.575	**0.608**	0.635	−0.61	−1.18	−0.69	−0.34
Caption length (words)	21.0	16.4	17.0	19.0	11.3	+2.40	+1.30	+1.13	+1.16

Table 2. Two-stage review of 50 sampled captions per model. Last row: inaudible phrases among the model's reference-unsupported phrases.

	GPT	Gemini	Qwen	AF3
Stage 1: reference-agreement category				
1–3 Aligned	15	8	21	**26**
4 Partial, missing detail	10	6	5	5
5 Partial, added detail	10	12	9	9
6 Weak or mismatch	15	24	15	10
Stage 2: audio-grounded rating				
1 Accurate	**47**	23	25	27
2 Minor errors	3	18	20	16
3 Major error	0	9	5	7
Category 6 with major error	0	6	1	1
Inaudible / ref.-unsupported	1/59	34/80	20/58	19/65

tion. The intervals are wide, so this analysis can neither confirm nor exclude useful discrimination.

5. DISCUSSION

Reference agreement and acoustic correctness are different evaluation targets, and in the reviewed sample they diverged. GPT-Audio-1.5 illustrates the divergence rather than a general superiority: it ranks third or fourth on every text metric, yet 47 of its 50 sampled captions were rated accurate. Its departures from the references were mostly of kinds the rubric does not count as errors: 10 of its captions omit reference content, and 55 of its 59 reference-unsupported phrases were audible. Omitting a secondary sound and adding audible detail lower reference agreement but not acoustic correctness, whereas misnaming or inventing the principal source lowers both; reference-based scores cannot tell these cases apart.

The human baseline should be read in the same light: that every model falls below it means that the models write differently from Clotho's annotators, not that their captions are less correct. Longer captions and audible content missing from the references are both associated with the gap, although neither was isolated as a cause.

Audio-aware metrics are at best a partial remedy. CLAP similarity is the only metric on which a model was not significantly below the human baseline, and its model order came closest to the audio review, but neither CLAP nor MACE showed significant within-recording concordance with the audio ratings. For LALM captions we therefore recommend reporting audio-aware metrics alongside reference-based ones, and checking a sample of captions against the audio before reading score differences as differences in acoustic correctness.

6. LIMITATIONS

Measurement reliability. The audio ratings are an operational assessment by a single reviewer, not an independently established ground truth, and the reviewer had read the references before listening. *Statistical precision.* The audio review covers 50 recordings: the order of Audio Flamingo 3, Qwen2.5-Omni-7B and Gemini 3.8 Flash in it is not stable (33% of resamples), and the concordance analysis (156 pairs from 42 recordings) has little power, so its null result does not show that metrics cannot track acoustic correctness. *Generalizability.* Each model was run with one prompt and one decoding setting, overlap of the API models' undisclosed training data with Clotho cannot be ruled out, and the findings are limited to Clotho's everyday sounds.

7. CONCLUSION

Agreement with human references and correctness about the recording are distinct properties of LALM captions. On Clotho v2.1, the reference-based ranking of four LALMs differed from a single-reviewer audio-grounded assessment, and 56 of the 64 sampled captions with weak reference agreement had no major acoustic error. Reference agreement should therefore not be taken as sufficient evidence that a caption is correct about the recording.

Availability. Code and data: `https://github.com/anandaswarup/lalm-acoustic-caption-eval`.

8. ACKNOWLEDGMENTS

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9. COMPLIANCE WITH ETHICAL STANDARDS

This study uses the publicly available Clotho dataset and captions generated by AI models. No human participants were recruited, and no ethical approval was required.

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